

ASSIGNMENT



Inspiring Excellence

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Assignment No: 2

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Answer to the Question No. 1

Opcode is a significant part of the machine code. It specifies the operation that the instruction performs. It tells the processor about the operation to be executed.

The instruction format and the control signals that needs to be activated for that particular operation is determined by opcode. Opcode is generally 7-bit in size.

Answer to the Question No. 2

We need a separate type of instruction for store operations in RISC-V.

A store (S-type) needs a base address register rs1 and a register

containing data to be stored which is rs2. It doesn't require any destination register like the ones in I type. thus it does not match with the I-type format.

So, RISC-V introduces S-type where the destination field gets replaced by a 12-bit immediate which is then split.

Answer to the Question No. 3

$$a = [(A[B[3]] + B[24] - 12) * 28] / 256$$

$$A \rightarrow X_{20}$$

$$B \rightarrow X_{21}$$

$$a \rightarrow X_{22}$$

$$PW \rightarrow 64 \text{ bits}$$

(i) Loading B[3]

ld x5, 24(x2)

(ii) Loading A[B[3]]

slli x6, x5, 3

add x6, x6, x20

ld x7, 0(x6)

(iii) B[24]

ld x28, 192(x2)

(iii) $(A[B[3]] + B[24] - 12)$

add x9, x7, x28

addi x9, x9, -12

(iv) $2^5 - 2^2 = 28$

~~stli~~ x10, 28

sh slli x18, x9, 2

slli x19, x9, 5

sai x20, x19, x18

(v) Divide by 256

~~srl~~ ~~x20~~,

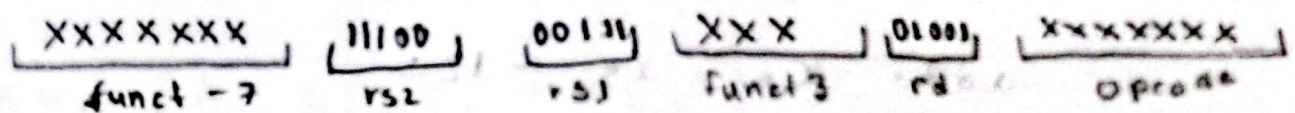
srl x20, x20, 8

(vi) Store the result in a

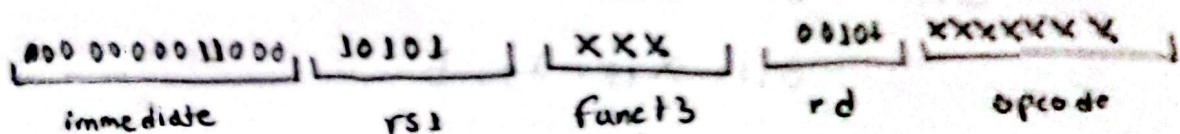
addi x22, x20, 0

(b)

add x9, x7, x23 - R (Type)



ld x5, 24[x21] - I (Type)



$sd\ x28, 102[x21]$

$102_{10} \rightarrow 11000000_2$

12 bits $\rightarrow 000011000000$

0000110	1100	1010	XXX	00000	XXXXXXXX
imm [11:5]	rs2	rs1	funct3	im [4:0]	opcode

Answer to the Question No. 4

In case of load instruction [funct3] specifies the size and signedness of the data being loaded. So, with different funct3 we get lb, lh, lw, dd, lbu, lhu etc.

Answer to the Question No 5

In S-type we need fields for $rs1$, $rs2$ and 12 bit immediate. The instruction must fit into 32 bits and keep the register field aligned. So, RISC-V splits the 12-bit immediate into:-

$imm[11:5]$ & $imm[4:0]$

(before)

(after)

As a result they keep the same total width and similar field positions as other formats.